

Python String interview questions

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Ques. Python Strings?

- Once a string object has been created, it cannot be changed. "Modifying" that string creates a whole new object in memory.
- 'hello' is the same as "hello".

Ques. Assign String to a Variable?

```
a = "Hello"
print(a)    # Output:- Hello
```

```
message = 'This is a string in Python'    # This is a string in Python
message = "This is also a string"         # This is also a string
message = "It's a string"                 # It's a string
message = 'mohit is a "good" boy'         # mohit is a "good" boy
```

Ques. Multiline Strings?

- You can assign a multiline string to a variable by using three quotes:

```
a = """Lorem ipsum dolor sit amet,
consectetur adipiscing elit,
sed do eiusmod tempor incididunt
ut labore et dolore magna aliqua."""
print(a)
```

Output:-

```
Lorem ipsum dolor sit amet,
consectetur adipiscing elit,
sed do eiusmod tempor incididunt
ut labore et dolore magna aliqua.
```

Ques. Check String?

- The **in** operator will return **True** if a string contains a substring.
- we can use the keyword **in**.

```
txt = "The best things in life are free!"  
print("free" in txt)    # Output:- True
```

- Use it in an if statement:

```
txt = "The best things in life are free!"  
if "free" in txt:  
    print("Yes, 'free' is present.")
```

Output:- Yes, 'free' is present.

- Check **if NOT**

```
txt = "The best things in life are free!"  
print("expensive" not in txt)    # output:- True
```

Ques. Slicing Strings?

- We can return a range of characters by using the slice syntax.
- Specify the start index and the end index, separated by a colon, to return a part of the string.

```
b = "Hello, World!"  
  
print(b[2:5])    # Output:- llo  -- Slice from the start position and end position  
print(b[:5])     # Output:- Hello  -- Slice From the Start: Get the characters from  
the start to position (5 not included)  
print(b[2:])     # Output:- llo, World! -- Slice To the End: Get the characters  
from position 2, and all the way to the end.  
print(b[-5:-2])  # Output:- orl  -- Negative Indexing: Get the characters from  
position 2, and all the way to the end.  
print(b[:-1])    # Output:- Hello, World
```

Change/Modify String

Ques. Modify Strings?

- **Upper Case:-** it method returns the string in upper case:

```
a = "Hello, World!"  
print(a.upper())    # Output:- HELLO, WORLD!
```

- **Lower Case:-** It method returns the string in lower case:

```
a = "Hello, World!"  
print(a.lower())    # Output:- hello, world!
```

- The **replace()** method replaces a string with another string.

```
a = "Hello, World!"  
print(a.replace("H", "J")) # Output:- Jello, World!
```

- The **split()** method splits the string into substrings if it finds instances of the separator

```
a = "Hello, World!"  
b = a.split(",")    # Output:- ['Hello', ' World!']
```

- **strip() method:-** The strip() method removes any whitespace from the beginning or the end.

```
b = "    Hello, World!    "  
print(a.strip())    # Output:- Hello, World!
```

String Concatenation

- To concatenate, or combine, two strings we can use the **plus(+)** operator.

```
a = "Hello"  
b = "World"  
c = a + b  
print(c)
```

Output:- HelloWorld

- To add a space between them, add a " "

```
a = "Hello"  
b = "World"  
c = a + " " + b
```

```
print(c)
```

Output:- Hello World

Format Strings

String Format:- As we learned in the Python Variables chapter, we cannot combine strings and numbers like this.

```
age = 36
txt = "My name is John, I am " + age
print(txt)
```

Output:-

Traceback (most recent call last):

File "demo_string_format_error.py", line 2, in <module>

```
    txt = "My name is John, I am " + age
```

TypeError: must be str, not int

we can combine strings and numbers by using the `__format__` method!

The `format()` method takes the passed arguments, formats them, and places them in the string where the placeholders `{}` are:

```
age = 36
txt = "My name is John, and I am {}"
print(txt.format(age))
```

output:- My name is John, and I am 36

The `format()` method takes unlimited number of arguments, and are placed into the respective placeholders:

```
quantity = 3
itemno = 567
price = 49.95
myorder = "I want {} pieces of item {} for {} dollars."
print(myorder.format(quantity, itemno, price))
```

output:- I want 3 pieces of item 567 for 49.95 dollars.

You can use index numbers `{0}` to be sure the arguments are placed in the correct placeholders:

```
quantity = 3
itemno = 567
price = 49.95
myorder = "I want to pay {2} dollars for {0} pieces of item {1}."
print(myorder.format(quantity, itemno, price))
```

Output:- I want to pay 49.95 dollars for 3 pieces of item 567

Ques. What is an f-string and how do you use it?

New in python 3.6, f-strings make string interpolation really easy. Using f-strings is similar to using format(). and F-strings are denoted by an f before the opening quote.

```
name = 'Mohit'  
food = 'Tea'  
print(f'Hello. My name is {name} and I like {food}.')
```

Output:- Hello. My name is Mohit and I like Tea.

Ques. Escape Characters?

- An escape character is a backslash \ followed by the character you want to insert.

```
txt = "We are the so-called \"Vikings\" from the north."
print(txt)
```

Output:- We are the so-called "Vikings" from the north.

```
# Single Quote
txt = 'It\'s alright.'
print(txt)
```

Output:- It's alright.

```
# Backslash
txt = "This will insert one \\ (backslash)."
print(txt)
```

Output:- This will insert one \ (backslash).

```
# New Line
txt = "Hello\nWorld!"
print(txt)
```

Output:-

Hello
World!

```
# Carriage Return
txt = "Hello\rWorld!"
print(txt)
```

Output:-

Hello
World!

```
# Tab
txt = "Hello\tWorld!"
print(txt)
```

Output:- Hello World!

```
# Backspace
txt = "Hello \bWorld!"
print(txt)
```

Output:- HelloWorld!

```
# Octal value
txt = "\110\145\154\154\157"
print(txt)
```


Output:- Hello

```
# Hex value
txt = "\x48\x65\x6c\x6c\x66"
print(txt)
```

Output:- Hello

Ques. How would you confirm that 2 strings have the same identity?

The **is** operator returns True if 2 names point to the same location in memory.

```
animals          = ['python', 'gopher']
more_animals     = animals

print(animals == more_animals) #=> Output:- True
print(animals is more_animals) #=> Output:- True

-----

animals          = ['python', 'gopher']
even_more_animals = ['python', 'gopher']
print(animals == even_more_animals) #=> Output:- True
print(animals is even_more_animals) #=> Output:- False
```

Ques. Check if each word in a string begins with a capital letter?

The **istitle()** function checks if each word is capitalized.

```
print( 'The Hilton'.istitle() ) #=> True
print( 'The dog'.istitle() ) #=> False
print( 'sticky rice'.istitle() ) #=> False
```

Ques. When would you use rfind()?

- **rfind()** is like find() but it starts searching from the right of a string and return the first matching substring.

```
story = 'The price is right said Bob. The price is right.'
story.rfind('is')
```

Output:- 39

Ques. Count the number of a specific character in a string?

- **count()** will return the number of occurrences of a specific character.

```
str = 'The first president of the organization..'
print(str.count('o'))    # Output:- 3
print('The first president of the organization..'.count('o')) # Output:- 3
```

Ques. find the length / Count number of characters in a string

- To get the length of a string, use the **len()** function.

```
a = "Hello, World!"
print(len(a))

Output:- 13
```

- Without function

```
na = input("Enter a string: ")
a = 0
for i in na:
    if i!=" ":
        a = a+1
print(a)

Output:-
Enter a string: mohit saxena
11
```

```
string = "My Name is Mohit Saxena";
count = 0;

for i in range(0, len(string)):
    if(string[i] != ' '):
        count = count + 1;
print("Total number of characters in a string: " + str(count));

Output:- 19
```

Ques. Remove vowels from a string?

```
string = 'Hello mohit saxena'
vowels = ('a','e','i','o','u')
new_string = ''
```

```

for ele in string:
    if ele not in vowels:
        new_string = new_string + ele
print(new_string)

# Using comprehension
print(''.join([c for c in string if c not in vowels]))

```

Output:- Hll mht sxn

Ques. Extract numbers from string?

```

new_string = "Germany26China47Australia88"

emp_str = ""
for m in new_string:
    if m.isdigit():
        emp_str = emp_str + m
print("Find numbers from string:", emp_str)

```

Output:- Find numbers from string: 264788

```

# 2 Example
new_str = "Micheal 89 George 94"

emp_lis = []
for z in new_str.split():
    if z.isdigit():
        emp_lis.append(int(z))

print("Find number in string:", emp_lis)

```

Output:- Find number in string: [89, 94]

Ques. Find repeated characters in a string python?

```

string = "Great responsibility";

for i in range(0, len(string)):
    count = 1;
    for j in range(i+1, len(string)):
        if(string[i] == string[j] and string[i] != ' '):
            count = count + 1;
            string = string[:j] + '0' + string[j+1:];

    if(count > 1 and string[i] != '0'):
        print(string[i]);

```

output:- r

```
e
t
s
i
```

Ques. program to count the frequency of each character?

```
# using "in" operater
str1 = input ("Enter the string: ")
d = dict()
for c in str1:
    if c in d:
        d[c] = d[c] + 1
    else:
        d[c] = 1
print(d)

Output:-
Enter thestring: Hhello
{'H': 1, 'h': 1, 'e': 1, 'l': 1, 'l': 1, 'o': 1}
Enter the string: Hello My name is Mohit Saxena
{'H': 1, 'e': 3, 'l': 2, 'o': 2, ' ': 5, 'M': 2, 'y': 1, 'n': 2, 'a': 3, 'm': 1,
'i': 2, 's': 1, 'h': 1, 't': 1, 'S': 1, 'x': 1}
-----

# Use of "get()" function
```

Ques. Looping Through a String?

```
for x in "Mohit":
    print(x)

Output:-
M
o
h
i
t
```

Ques. Reversed the String?

- Using slice method and through function

```
txt = "Hello World"[::-1]
print(txt)
```

```
str = input("enter the string: ") # sky is blue
split = str.split()
split = split[::-1]
finalstr = " ".join(split)
print(finalstr)
```

Output:- blue is sky

- using reversed() function

```
string = "Hello, World!"
print("".join(reversed(string)))
```

Output:- !dlrow ,olleH

- Using Function

```
def my_function(x):
    return x[::-1]

mytxt = my_function("Hello World")

print(mytxt)
```

Output:- dlrow olleH

- without function

```
na = input("Enter a string: ")
count = 0
blank = ""

for i in na:
    count = count+1
for i in range(1, count+1):
    blank = blank+na[count-i]
print(blank)
```

Output:-
Enter a string: mohit
tihom